

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Study of {space / stars / galaxies} (1) don't accept planets For the 2 nd mark either: At {different / various} {wavelengths / frequencies / energies} of the em spectrum OR Of various regions of the em spectrum	2			2		
	(b)		W in base units $\rightarrow \text{kg m}^2 \text{s}^{-3}$ or $\text{J} \rightarrow \text{kg m}^2 \text{s}^{-2}$ or $\text{N} \rightarrow \text{kg m s}^{-2}$ (1) Units of $\sigma = \text{kg s}^{-3} \text{K}^{-4}$ (1)	1	1		2	2	
	(c)	(i)	Use of $T = \frac{W}{\lambda_{\text{max}}}$ (re-arrangement and substitution) i.e. $T = \frac{2.9 \times 10^{-3}}{674 \times 10^{-9}}$ (1) $T = 4303 \text{ K}$ (1) unit mark		2		2	2	
		(ii)	Red / orange	1			1		
	(d)		Correct substitution into $P = A\sigma T^4$ i.e. $P = 5.67 \times 10^{-8} \times 4\pi \times (1.39 \times 10^{10})^2 \times (4303)^4$ ecf on T Or $P = 4.72 \times 10^{28} \text{ [W]}$ seen (1) Recall $I = \frac{P}{4\pi D^2}$ (1) Alternative 1 - Substitution and rearrange i.e. $D = \sqrt{\frac{4.72 \times 10^{28} \text{ ecf}}{4\pi \times 3.09 \times 10^{-8}}}$ (1) $D = 3.49 \times 10^{17} \text{ [m]}$ (1) $D \text{ in light years} = \frac{3.49 \times 10^{17}}{9.46 \times 10^{15}} = 36.8[5 \text{ ly}]$ so claim correct ecf on D (1)			5	5	5	

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			Alternative 2 – Substitution and rearrange i.e. $I = \frac{4.72 \times 10^{28} \text{ecf}}{4\pi \times (37 \times 9.46 \times 10^{15})^2} \text{ (1)}$ = 3.07 × 10⁻⁸ [W m⁻²] (1) Compare with 3.09 × 10⁻⁸ [W m⁻²] so claim correct or valid conclusion if ecf used Alternative 3 – Substitution and rearrange i.e. $P = 3.09 \times 10^{-8} \times 4\pi \times (37 \times 9.46 \times 10^{15})^2 \text{ (1)}$ = 4.72 × 10²⁸ [W] Compare intensities so claim correct or valid conclusion if ecf used (1) To summarise: Correct substitution into Stefan's law (1) Quoting or using $I = \frac{P}{4\pi d^2}$ (1) Correct substitution and algebra (1) Correct almost final value (1) Comparison and valid comment (1)						
			Question 7 total	4	3	5	12	9	0